

Additively manufactured heat exchangers: a review on opportunities and challenges

Seyed A. Niknam^{1,*}, Mehdi Mortazavi², Dongsheng Li³

¹ Department of Industrial Engineering, Western New England University, Springfield, MA

² Department of Mechanical Engineering, Western New England University, Springfield, MA

³ Advanced Manufacturing LLC, East Hartford, CT 06032

Key words: heat exchanger, additive manufacturing, selective laser melting, pin fin arrays, surface roughness, friction factor

Abstract

Heat Exchangers (HXs) are crucial engineering components for thermal management. The broad range of applications and the current limits of traditional manufacturing techniques has directed designers and researchers to utilize additive manufacturing (AM) technologies to open a new chapter in HXs design and development. AM has created a world of opportunity to redesign HXs in pioneering forms, shapes, and sizes. In other words, AM technologies provides significant opportunities to increase surface area to volume ratio in heat transfer devices. On the other hand, there are certain challenges to overcome in order to fully grasp the benefits of AM technologies. This thorough review study is different from other published review as it intends to highlight the challenges and opportunities in design and manufacturing of HXs. In addition, part of this study is dedicated to investigate the experimental characterization of additively manufactured HXs. The significant details associated with experimental characterization are normally overlooked in review studies.

Contents

1. Introduction.....	2
2. HX Design using AM	3
2.1 Design opportunities	4
2.2 Important design features.....	6
2.3 Geometrical optimization.....	8
3. HX fabrication using AM	9
3.1 AM Materials	10
3.2 Cold Spary	13
3.3 Fabrication opportunities and challenges.....	14
4. Challenges in Experimental characterization.....	15

* Corresponding Author, seyed.niknam@wne.edu, phone: 413-782-1354, Fax: 413-796-2116

4.1 Surface roughness	Error! Bookmark not defined.
4.2 Heat Transfer	Error! Bookmark not defined.
5. Summary and conclusions	20
References	22

1. Introduction

Heat Exchangers (HXs) are crucial components for thermal management. Exchanging heat between two fluids is the primary use of HXs, which creates applications in a broad range of engineering systems such as power plants, oil refineries, air conditioning, automotive, oil coolers in aircrafts, high-temperature resistojets, compact electronic cooling, chemical processes, waste heat recovery devices (e.g. recuperator) in furnaces and hydrogen plants, energy storage systems, and energy conversion systems such as refrigeration cycles [1–6].

For all practical purposes, the overall system reliability depends heavily on effective heat transfer [7]. There are distinct types of HXs based on process and operating environment [1] with the primary goal of maximum heat transfer as well as lowest pressure drop, i.e. pumping power consumptions [3]. Shell and tube HXs are the most common type of HXs for liquid-liquid applications. There are some concerns about the size and space limitation for this type of HXs, but innovations are underway [8, 9]. The other prevalent type of HXs is plate-and-frame which are normally applied for two liquids. Various patterns of plate and layout are being used in this type of HXs [4]. In essence, the HXs specific application dictates its final type [10]. There are also various forms of passive heat transfer devices such as heat pipe (HP) and heat sinks (HS). Heat pipes (HP) are two-phase devices that transfer heat at high rate based on thermal conductivity and phase change of the working fluid. There are distinct mechanisms in HPs to help return condensate flow to evaporator, which include gravity, wicking effect, magnetic field, and centrifugal force [11].

Recently, there has been growing interests in redesigning traditional HXs, which dates back to decades. Besides, there is an increasing trend in developing small-size HXs, particularly for compact electronic devices [7]. In this way, additive manufacturing (AM) technologies has been considered as an effective prescription to remove the limits of traditional manufacturing techniques and opens a new chapter in HXs design and development. AM refers to a class of technologies based on layer-by-layer fabrication using 3D model data [12]. There are several examples of novel HX design in the literature that sound very challenging, if not impossible, to be fabricated through traditional manufacturing processes. AM has created a world of opportunity to redesign HXs in pioneering forms, shapes, and sizes. In other words, AM is able to facilitate the need for higher surface area to volume ratio in heat transfer devices. Moreover, AM is a new opening for devices such as microchannel HXs, which could not find broad applications due to high cost of conventional manufacturing methods.

There are many comprehensive studies in the literature explaining various AM technologies. For the sake of conciseness, this study will not provide the details for various AM technologies and readers are referred to Ref. [13, 14]. The main metal technologies considered in this study are powder bed fusion (PBF) and powder feed AM [15]. Generally, powder additive manufacturing methods are divided into two categories based on powder delivery system: powder

feed and powder bed. In powder bed AM category, powder was delivered first into one layer of bed spread either by roller, scraper, or blade. Then the powder was solidified into desired shape by heat source from laser, electron beam, UV source, or other electromagnetic fields. The powder bed AM fabrication methods include the following commonly used printing techniques: Direct metal laser sintering (DMLS), Electron beam melting (EBM), Selective heat sintering (SHS), Selective laser melting (SLM) and Selective laser sintering (SLS). The high-power electron beam would lead to parts with better mechanical strength compared to other metal AM technologies.

On the contrast, in powder feed AM category the powder is delivered and solidified simultaneously layer by layer. Powder feed method is faster and more cost efficient. Powder bed method is generally more precise with higher resolution. The most well-known powder feed AM technique is direct energy deposition (DED) where metal feedstock will be deposited directly onto a build tray or an existing part by an energy source, building up free form three-dimensional parts [12]. The material is deposited as continuous flow of powders or by means of wire feedstock. Application of DED in heat exchanger and related functional components has been expanded into two directions: one is on large scale and fast rate with the goal to decrease cost and increase efficiency; the other one is on expand design and materials freedom, increase resolution and decrease roughness. Both two directions are critical to success fabrication of lighter, cheaper and easier made rocket engine for NASA [16–19].

There is another form of powder feed AM process known as cold spray, which applies kinetic energy to establish solid-state bonds. Cold-gas dynamic spraying, or simply cold spraying (CS), is a solid-state powder deposition technology in which powder consolidation is made by high velocity impact of accelerated particles on a build surface [20]. This fabrication technique will be elaborated in Section 3.2. In addition, although laminated object manufacturing (LOM) found some use for HXs manufacturing, this AM technique is not as mature as SLM and DED in providing accurate and high-quality parts.

This review study intends to highlight the challenges and opportunities in design and manufacturing of HXs. Section 2 is dedicated to the design opportunities and challenges in fabricating HXs using AM technologies. Section 3 evaluates opportunities and challenges offered by AM from manufacturing point of view. In section 4, the significant insights in experimental characterization of additively manufactured HXs are extensively considered. Section 5 provide the conclusions and the future directions.

2. HX Design using AM

There is always a definitive need for compact and efficient HXs. This implies a demand for optimal heat transfer in a smaller space with less material, i.e. higher heat transfer and reduced mass to volume ratio [5]. Heat transfer enhancement for single-phase devices comes normally with an increase in pressure drop between the channel inlet and outlet [5], especially in micro-channels [1]. In essence, low pressure drops ensure low pumping power requirements [7]. Thus, design goals for HXs can be summarized as maximizing the ratio of heat transfer to pressure drop [21]. In addition, for two-phase flow devices, understanding the flow pattern is critical. Several studies considered two-phase flow in additively made heat pipes and porous lattice structures [22–26].

2.1 Design opportunities

Cost-effective fabrication of complex and compact designs through AM has led many researchers to embrace this significant opportunity and explore new HX designs such as bio-inspired wavy microchannels [21], miniature water-cooled HX [27], innovative manifold microchannel [28, 29], topology-optimized HX [7], tree-like distributor for a multi-channel HX-reactor [30], innovative Pitot HS [31], airfoil pin fins [32], internally finned tubes [33], finless tube bundle with novel shapes [34], and webbed-Tube HX [35] to name a few. Figure 1 shows some examples of novel HX design fabricated via AM. It is important to note that some of the new HX designs have not been tested in actual applications. For instance, Cárdenas et al. [36] used hexagonal mesh for the cross-sectional area for an air-to-air HX. Although it was not proved experimentally, the authors stated the new design may result in significant volume reduction per unit of energy transfer [36]. Saltzman et al. [1] presented a new design of aircraft oil cooler which required major geometry changes for manufacturability. A novel HS design presented in [37] and fabricated via powder bed fusion, could reduce the volume and weight by 98% and 94%, respectively. Gerstler and Erno [38] came up with a new design of multi-furcating HXs for fuel-cooled oil cooler applications using four different materials and SLM technology. While they achieved 66% lower weight and 50% lower volume, the performance of additive HXs was found to be equivalent to a conventional design [38]. Bernardin et al. [39] designed a compact twisted tube shell HX fabricated via SLM. The performance analysis was merely done numerically. Comparing to the traditional round tube HXs, this type of swirl design is a better design approach because it generates a secondary flow within the axial flow and leads to reduction of the boundary layer.

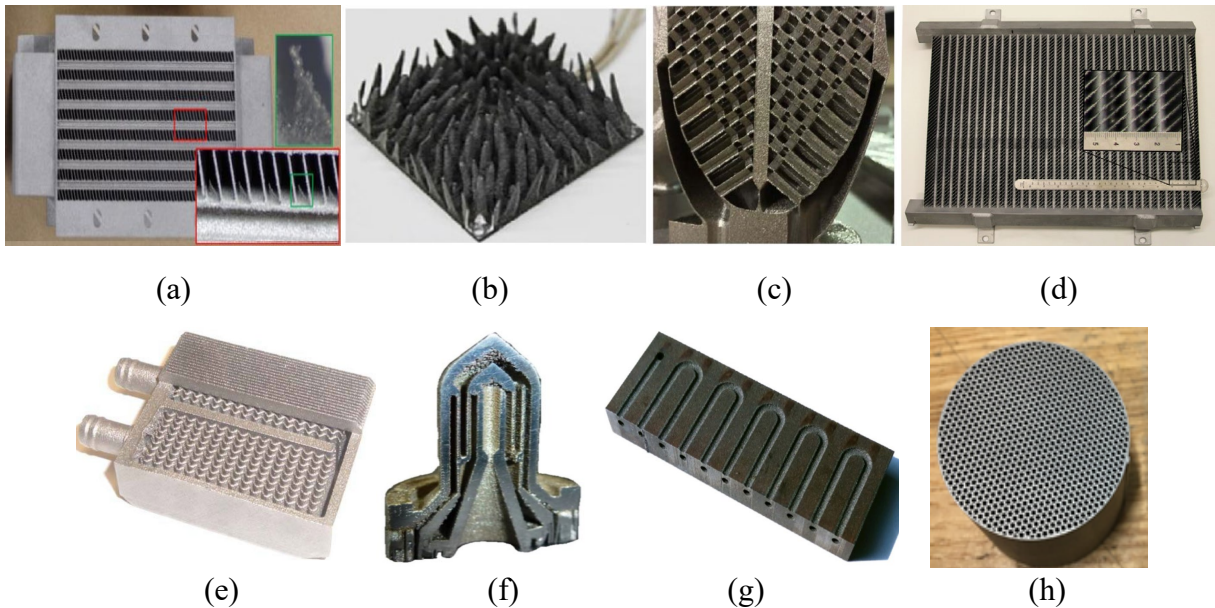


Figure 1. Examples of novel design of HX fabricated via AM; (a) enhanced aircraft engine oil coolers [1], (b) topology-optimized heat sink with variable pin fin [7], (c) multi-furcating heat exchanger [38], (e) miniature water-cooled HX [27], (f) sectioned view of a novel resistojet heat exchanger [2], (g) sectioned oscillating heat pipe with circular mini channels [40], (h) Cross section of a new conversion reactor [8].

Microchannel HXs seem promising because of their high heat transfer rates. In effect, novel designs of manifold-microchannel HXs reduce the microchannel flow length, and therefore, resolve the issue of high pressure drop which is intrinsic to narrow flow channels [3]. In an attempt to develop more affordable dry cooling for power plants, Arie et al. [41] studied three air-water manifold-microchannel HXs made by aluminum alloy (AlSi10Mg), stainless steel (SS17-4), and titanium alloy (Ti64). The water side of HX comprised rectangular channels in a crossflow configuration. The authors considered the temperature difference between the inlet water and the inlet air at the manifold-microchannels (ΔT). They showed Ti64 prototype could improve gravimetric heat transfer density by 25%-30% compared to wavy fins and plain plate fins. In this respect, the gravimetric heat transfer can be defined as $Q/m\Delta T$, where m is the heat exchanger mass and Q is heat flow rate.

Similar to HXs, chemical reactors are also receiving the benefits of AM-enabled designs. For example, multi-furcating configuration was explored by Gerstler and Erno [38]. The complex multifurcated design decreased weight 66% and shrank the volume more than 50% when keeping the same heat transfer efficiency and pressure drop as the traditional design. More elaborate architectures, such as mixing structures, residence time channels, separation units, and interfaces for in-line analysis, have been incorporated into within microreactors and millireactors. One exemplary work has been performed by research group of Gutmann et al. [42], in which a reactor for continuous difluoromethylations using fluoroform was designed and additively manufactured using stainless steel 316L (see Figure 2). They claim their 3D printed device is the first metal reactor in the world for synthesis. Shortcut holes were designed at every turn for easy removal of powder. After the channels were cleared and the reactor was cleaned with ultrasound, the holes were closed by laser beam welding and 1/16" stainless steel tubes were attached onto the inlets and outlets. The tubes were finally equipped with standard 1/16" fittings to allow simple integration with peripheral flow equipment.

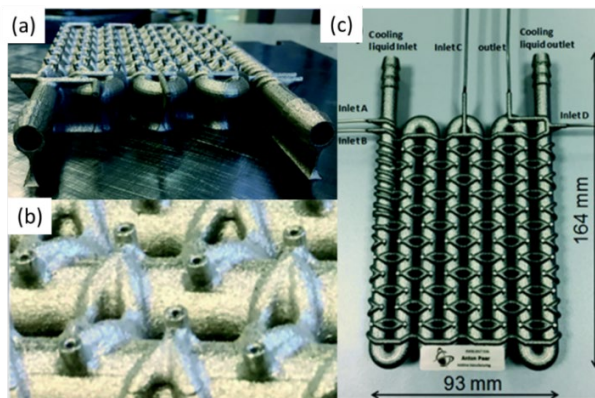


Figure 2. The additively manufactured chemical reactor. (a) Top view, (b) Shortcut holes incorporated at every turn. (c) 3D perspective showing four chemical inlets and cooling liquid inlet and outlet.

2.2 Important design features

This section highlights important HXs design features which can be fabricated using AM technologies but relatively challenging to be fabricated through conventional methods. The new or enhanced design features - such as microgrooves, louver features, and vortex generators - are expected to assist in increasing the surface to volume ratio as well as reducing the volume to weight ratio to some extent. In this section, certain significant features enabled by AM with potential for performance enhancement are considered. In this respect, vortex generators has been mentioned by a handful of researchers as an effective design tool for enhancing heat transfer without severe effect on air-side pressure drop [1]. Vortex generators are known as a type of fin which disturb boundary layer flows [5]. Aris et al. [43] applied SLM technology to fabricate active delta wing vortex generators made from shape memory alloy materials for microelectronics applications. In their experimental investigation, the authors in [43] demonstrated heat transfer enhancement of up to 90% and 7% pressure loss by the single wing vortex generators mounted on a heated rectangular channel surface. In addition, internal rifling was used by Tiwari et al. [3] to improve heat transfer in a tubular heat exchanger. Internal rifling would disturb the temperature gradients near the tube surface. Moreover, wavy channels have been a subject of study in order to enhance heat transfer and reduce the size and pressure drop in [21], [44], and [45]. One can expect higher heat transfer in wavy channels compared to straight channels. Kirsch and Thole demonstrated that longer wavelength channels provide higher heat transfer than short wavelength channels. High pressure drop was reported in short wavelength channels [45].

The most noteworthy trend in redesigning HXs is the elimination of conventional fins. Although fin arrays can provide compact design and reduction in thermal resistance, one should expect increase in viscous resistance and higher manufacturing costs. Traditionally, fins are stamped and then brazed to the base plate. In high pressure HXs, brazing is replaced by diffusion welding. Continuous fin arrays can be divided into two major classes: plate or wavy. It is evident that the heat transfer performance depends heavily on fin size, number of fin layers, and type of fins. Nevertheless, reinforced by AM technologies, there is a tendency to transform traditional continuous fin to discrete pin fin arrays. The notable properties of pin-fin arrays are as follows: high volumetric heat transfer [46, 47], augmented turbulent mixing [4], potential increase in pressure drop [48], and durability concerns due to high mechanical and thermal stresses.

Heat transfer in pin-fin arrays is a function of pin geometries such as pin size and shape, fin array parameters such as pin spacing, roughness level, and shroud clearance [49]. Interestingly, Kirsch and Thole [10] showed similar heat transfer performance for two pin-fin arrays in which one array has six times more pins. Similar results were reported by Alsulamin et al. for manifold-microchannel heat exchanger [50]. In essence, understanding tradeoff between design complexity and performance would be very helpful for weight reduction as well as manufacturing cycle time. In addition, based on the thermo-hydraulic performance comparison presented in [5], it seems additively fabricated pin fin arrays provide higher friction factor augmentation compared to that of wavy microchannels.

Considering the pin geometries, Wong et al. concluded that elliptical fins, made via SLM, have better heat transfer per unit pressure drop compared to rectangular and cylindrical pin [51]. They also examined lattice shaped fins and indicated no promising heat transfer improvement despite larger heat transfer surface area [51]. In addition, it is important to point out that higher

surface area to volume ratios result in additional viscous resistance. Ho et al. [32] suggested better thermal performance for their novel airfoil fins in which the heat transfer coefficient would rise if the angle attack (α) is increased to 10° . The authors in Ref. [32] proved experimentally that airfoil pin fins provide superior forced convective heat transfer compared to the circular geometry (See Figure 3).

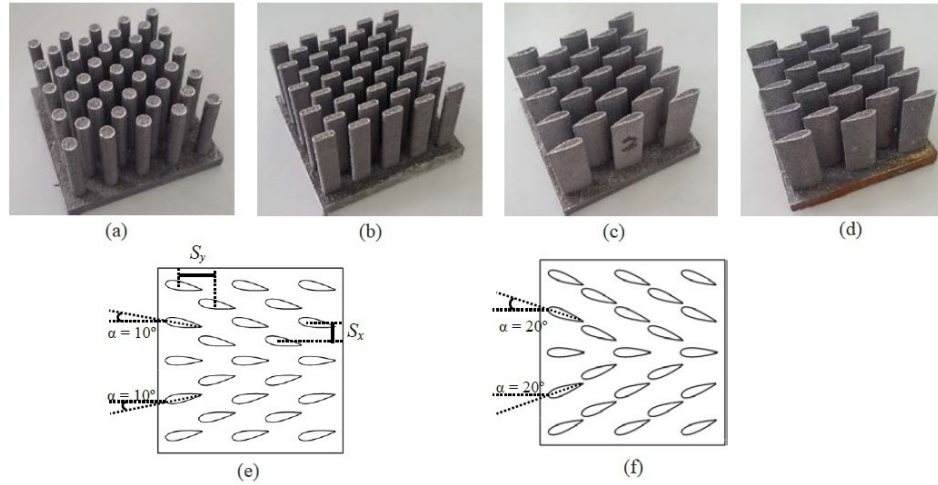


Figure 3. Top views of various pin fin arrays studied by Ho et al. [32]; (a) circular, rounded rectangular, (c) airfoil shaped fins NACA 0024 at ($\alpha = 0^\circ$), (d) airfoil shaped fins NACA 4424 at, (e) NACA 0024 at ($\alpha = 10^\circ$, $S_y = 10 \text{ mm}$, $S_x = 5 \text{ mm}$), (f) NACA 0024 at ($\alpha = 20^\circ$)[32]

Dupuis et al. investigated tapered fin arrays with square, diamond and round profiles in the inline and staggered arrangements [47]. The authors concluded that inline configurations produce less pressure loss as well as less convective coefficients. The authors in [47] recognized a better overall efficiency for elliptical shaped fins compared to cylindrical fins. Square base pin fins also provided better performance than cylindrical fins [47]. The early flow separation due to a lower surface area is mentioned as the reason of poor performance in cylindrical pin fins. Moreover, Dupuis et al. stated that longer pin fins (with higher height to diameter ratio) are likely to transfer more heat [47]. In this work, the fins were produced on an aluminum substrate ($51 \times 51 \text{ mm}$) with the height of almost 1 mm . This fins arrangement can be properly fabricated using powder bed fusion technologies.

Fasano et al. [31] took into account Pitot tube effect and significantly enhanced passive heat transfer in an innovative HS design. This was done by contriving fin cavities which allow the secondary flows exit orthogonally to the main flow. Robinson et al. [52] introduced FINJETTM architecture as an array of fins with integrated microjets embedded in a water-cooled microchannel HS. Few researchers have studied tapered pin fin arrays [47, 53, 54]. While there is no significant change in the fluid flow, inline pyramidal pin fins showed better thermal performance compared to constant cross-section fins [54]. The recirculation pattern in the wake of fins was observed in tapered pin fin and introduced as the reason behind the superior thermal performance [53]. Wang et al. [55] investigated cooling experiments based upon single water droplet on pin fin arrays with various geometries fabricated by SLM. The arrays were designed with high, medium, and low

density. The authors concluded that fin density and clavate are important contributing factor in the droplet dynamics [55].

The SLM technology has been considered by researchers to design freeform HPs with innovative shape, optimized wick structure, and compact size to expose novel applications. Ameli et al. [23] applied SLM to produce sintered wick structures for their aluminum based heat pipe. The wick structures in that work are different in thickness, porosity, permeability and pore sizes [23]. Esarte et al. [22] came up with a novel loop heat pipe (LHP) for cooling purposes. The performance of LHP devices depend on the design of the evaporator and primary wick. Using SLM, the researchers in [22] optimized the wick performance by embedding internal passages into the primary wick and scaffold porosity of 17%. A couple of research groups have worked on pulsating heat pipe (PHP) which requires no capillary structures. Among them, Thompson et al. [56] introduced a novel design of compact flat-plate PHP. They employed the concept of “vent holes” into their HP in order to de-powder the internal long channels layer-by-layer. In a similar work, Ibrahim et al. [40] tested a multi-layered PHP with several working fluid and different operating orientations. In addition, Jafari et al. [57] highlighted the significance of interstitial fluid on thermal conductivity based on the experimental results on SS316L porous HPs.

The SLM technology also offers the opportunity of fabricating innovative porous structures to suit various applications [23, 58]. A solid matrix with an interconnected void is known as porous media [59]. The possibility of fabricating various porous cell topologies makes this area even more attractive for designers particularly for improving area to volume ratio (which enhance boiling heat transfer) and capillary scale two-phase heat transfer equipment. Nevertheless, for small capillaries, it is challenging to establish a practical trade-off of some conflicting factors such as permeability and thermal conductivity. Beside the design challenges, there are some manufacturing concerns including accurate scanning speed, heat source power, build orientation, powder adhesion, and design of the unit cell [23, 58, 60, 61]. In other words, both process parameters and design parameters need to be optimized to achieve a precise porous structure. This opens a new research direction in this area. Hence, further investigations are necessary to reveal the potentials of porous media for HX applications and characterize the thermal-fluid behavior. Ho et al. [24] investigated heat transfer in a SLM-fabricated saturated pool boiling which was redesigned with micro-cavity and micro-fin surfaces features. The experimental comparison with commercially available plain Al-6061 surface showed the positive effects of AM-enabled features on enhancing heat transfer performances. In similar studies [25, 26, 62], the researchers confirmed the significant effects of SLM-fabricated porous structures on heat transfer coefficients, hydrodynamic choking, and critical heat flux. In [26], two-phase study of porous metallic structures revealed higher pressure drop at adiabatic condition.

2.3 Geometrical optimization

There can be no doubt that AM has given a boost to topology optimization (TO). Unlike traditional parametric design approaches, TO is a numerical top-down approach that generate efficient geometry under a prescribed objective and constraints [7]. In traditional practices, optimization of heat transfer devises is performed by manipulating geometric parameters in commercial computational fluid dynamics (CFD) software packages. Reducing design cycle time and demanding less computational power are the potential benefits of TO. Techniques of TO can

be classified into three branches; (i) density-based method, (ii) Evolutionary Structural Optimization (ESO) methods, and (iii) Level-Set based Method (LSM) [63].

AM-based TO faces numerous challenges. In practice, TO outcomes include organic free-form surface with a large number of long overhang features, which creates the need of support structures. Research is currently underway to overcome these challenges [64]. For instance, Gaynor and Guest proposed a TO framework using overhang constraint to minimize the need for support materials [65]. Recently, more attention has been given to lattice structure topology optimization (LSTO) for its proven ability to maintain the original design and the limited need for support structures [66]. Also within this context, Cheng et al [63] applied LSTO with implicit boundary conditions and designed a cooling channel with enhanced performance.

Koga et al. [67] worked on topology-optimized cooling channels with application to small scale heat sinks used in compact electronic components. The authors recognized the practicality of TO methods to solve multi-objective and multi-physics problems, i.e. maximizing heat dissipation combined with minimizing pressure drop. Das and Sutradhar [68] worked on generating functionally graded porous structure assisted by multi-physics topology optimization for heat-dissipating structures. Bacellar et al. [34] presented shape optimization using automated computational fluid dynamics simulations for a small HX with finless tubes. Their optimized design outperformed the baseline microchannel HX and showed over 50% potential size reduction. The authors in [34] believe all design variables (not only tube shape) should be taken into account to achieve better performance. Recent studies in this field have focused on design optimization for two-fluid heat exchanger. This class of topology optimization involves two-fluid and one-solid distribution. In this regard, Tawk et al. [69] extended the density-based approach in two-dimensional examples for bi-fluid applications with the objectives of minimizing the pressure drop and maximizing the heat transfer. Kobayashi et al. [70] stretched the two-dimensional two-fluid studies into three dimensions under the constraint of a fixed pressure loss. A detailed discussion on topology optimization is out of the scope of the current study. The interested readers are referred to [64, 71] and the recent review study on fluid-based topology optimization [72].

3. HX fabrication using AM

This section intends to highlight the remarkable opportunities offered by various AM technologies to fabricate efficient heat transfer devices. It is important to point out there are many non-additive manufacturing methods to fabricate HXs, such as chemical etching and diffusion bonding. Some of these methods are versatile in using different materials and fabricate with high accuracy. However, the main objective of this section is to highlight the main advantages of AM over traditional manufacturing particularly for HX applications. AM technologies in general and SLM in particular have a great potential to fabricate practical HXs. There are several important considerations in this respect including cost (vs quantity) [73], complexity [50, 74], and microstructural properties [75]. In essence, the prevalence of metal AM technologies is contingent on overcoming a number of technical challenges such as lack of standards, limited material options, setting suitable process parameters, difficulties in characterizing thermophysical properties, and acquiring the desired surface texture. This section starts with material-related challenges.

3.1 AM Materials

HXs have been made using all classes of materials, i.e. metal, plastic, ceramic, and composites. In material selection for AM, one needs to deliberate mechanical, chemical and thermal properties [4]. However, for heat transfer devices, thermal conductivity and durability are the key factors in the material selection process [1]. It should be emphasized that applicable materials for metal AM applications are growing, albeit rather slowly. In essence, high temperature materials, such as ferritic steels, Ni-based alloys and ceramics, are more favorable for certain HX application [4]. This section starts with metallic materials which is classified to ferrous, nonferrous, and superalloys. Table 1 shows the popular material for additively manufactured HXs.

3.1.1 Metals

Ferrous alloys (e.g. SS347, SS316L, etc.) are low cost materials with an extensive application in heat transfer devices. In particular, ferrous alloys are considered in HX designs with substantial components thermal expansion. On the other hand, nonferrous alloys might be better candidates for additively manufactured HXs applications. For instance, Aluminum alloys are favorable in SLM applications because of significant properties such as high strength-to-weight ratio and high thermal diffusivity. Wong et al. [76] showed that AL6061 can outperform SS 316L in air-cooling HSs fabricated with various fin geometries through SLM. Ho et al. [32] used AlSi10Mg powder to fabricate a novel airfoil HS with application to integrated circuit boards. It should be noted that AlSi10Mg is an appropriate material for electronic cooling because of relatively high thermal conductivity and light weight. In addition, because of corrosion-resistant properties of magnesium, AlSi10Mg is more preferable than other aluminum alloys such as AlSi12.

Copper has excellent thermal conductivity, and therefore, it is in demand for HXs applications and internal cooling channels. However, metal AM technologies are not mature enough to fabricate copper AM parts. This is mainly due to low laser radiation absorption and high thermal conductivity which creates the need for higher energy input [77]. In essence, research is underway to enable SLM and laser-based-DED systems to fabricate reflective metal such as copper or aluminum. For example, Zhang et al. [78] successfully fabricated components of Cu-15Ni-8Sn alloy with near full density of 99.4%. This high-performance and corrosion-resistant copper alloy sounds suitable for HXs applications. Popovich et al. [77] investigated the SLM fabrication of Cu-Cr-Zr-Ti alloy and achieved high density of high density of 97.9%. In this alloy, Zr and Ti are added to improve hot ductility. To improve the mechanical properties, the authors in [77] suggested further process parameters optimization to reduce residual porosity. Moriya et al. [79] used the similar alloy, i.e. Cu-Cr-Zr, in a SLM fabrication system and successfully prototyped a combustion chamber inner liner for very high-temperature applications. Sing et al. [80] reported effective AM methods in fabricating Cu alloys, Cu composite, and embedded copper structures. Such AM methods would provide the essential platform for more novel designs of multi-metal HXs, particularly in the field of electronics/electronic cooling. In essence, multi-material HXs is a great opportunity offered by AM in which smooth transition between two materials would remove joining and welding [19, 81–83]. Bimetallic or multi-alloy configuration for channel wall nozzles has been preliminarily investigated in [19]. Hot wall and inner liner were C-18150 copper alloy

and the closeout and jacket were Monel 400 nickel superalloy. Radial and axial structures have been successfully manufactured by laser wire direct closeout and explosive bonding respectively.

Table 1. Popular metals for additively manufactured HXs with selected references. If a test coupon used by a research group in several publications, it was only cited once in the table.

Material	Powder Bed Fusion	Cold Spray
<i>Stainless Steel (SS)</i>	<i>SS 316 L</i> : [2][8][9][28][22][57][76][84][42] <i>SS 17-4</i> : [41][50] <i>SS 347</i> : [85]	<i>SS 304</i> : [86]
<i>Ti-6Al-4V (Ti64)</i>	[34][38][41][40][56][59][63][87]	
<i>Inconel 718</i>	[10][21][38][44][85][88][89][90][91]	
<i>Nitinol (Ti-50at% Ni)</i>	[43]	
<i>Cobalt Chrome</i>	[30][38][90]	
<i>Aluminum alloys</i>	<i>AlSi10Mg</i> : [1][6][26][27][31][32][41][55][90][92][93] <i>Al 6061</i> : [38][51][24][76][90] <i>Pure Al</i> : [33][23] <i>AlSi12</i> : [7][95]	<i>Pure Al</i> : [47][53][96]
<i>Copper</i>	[80]	[97]
<i>EOS Nickel Alloy HX</i>	[98]	

Superalloys are an important group of heat-resistant materials in fabricating HX particularly for aerospace applications. Although superalloys normally come with high cost and low thermal conductivity, they are valued for elevated temperature applications as they can retain certain properties. In addition, despite their high toughness, they may suffer from crack formation during welding [4]. Among superalloys, titanium alloys have gained dramatically increasing attention from AM community because of cost advantages with respect to machining and long lead times [13, 14]. Research groups particularly paid attention to Ti-6Al-4V for all metal manufacturing processes [99, 100]. This alloy has been investigated for AM-fabricated HXs [38] and heat pipes [40, 56]. Increase in heat transfer was reported in [87] for air-water manifold microchannel HX made by Ti-6Al-4V.

Nickel-based super alloys are also suitable materials for various HXs applications. For high temperature applications, Inconel 718 provides high strength and excellent creep behavior. It should be pointed out that AM technologies are seriously being considered for Inconel 718 and other nickel-based superalloys mainly because of the potential for reduced production steps and enhanced geometrical complexity [82]. Made by Inconel 718 through SLM, Zhang et al. [88] demonstrated a light-weight gas-to-gas manifold microchannel HX with 25% higher heat transfer density at 600°C. Seiya and Zhang [85] pointed out the importance of post-print material properties for couple of materials including Inconel 718. Seiya and Zhang [85] also highlighted the economic viability in using metal AM technologies.

AM technologies, particularly laser-based DED, have created an unprecedented opportunity for various types of functionally graded components. This offers a huge potential for designers to produce variable-property and cost-effective HXs by density control, dimensionality control, or mixing various materials [101]. Although the technologies related to functionally graded AM is still immature, it will emerge as a feasible performance modeling approach in the

near future. In actual practice, the literature suggests several examples of promising studies. For instance, Onuike et al. [102] investigated a functionally graded material (FGM) with transition from Inconel 718 to a copper alloy (GRCop-84). This FGM shows 250% increase in diffusivity and 300% increase in conductivity compared to pure Inconel 718 alloy [102]. A density-controlled functionally graded Inconel 718 was produced via SLM by Popovich et al [103]. This investigation reveals a substantial improvement in thermomechanical life of the functionally graded Inconel 718 after heat treatment [103].

3.1.2 Ceramics

In the past two decades, ceramic materials have gained increasing attention to be used for HXs mainly in high temperature applications [4]. Ceramic HXs for high pressure application need to use monolithic bonds, i.e. permanent bonds. Furthermore, non-monolithic bonds provide the chance to be de-bonded to fix or replace the damaged parts. AM technologies have faced numerous challenges for processing ceramic materials. In particular, direct deposition techniques impose some limitations to fabricate ceramics. The challenges include but are not limited to feedstock and sintering [104]. In this context, a pioneer study by Alm et al. [105], applied stereolithography (SLA) to fabricate the injection mold and produced a ceramic plate-and-frame HX. Scheithauer et al. [106] introduced lithography-based ceramic manufacturing as an alternative of SLA with advantages such as better component properties and higher resolution. Shulman and Ross [107] prototyped a compact ceramic HX using laminated object manufacturing (LOM). In this process, ceramic green tapes were laser cut and then precisely stacked to build a HX with internal features. They applied zirconia-toughened mullite (ZTM) which offers several benefits for high temperature HX including ease of densification and relatively low cost. The prototype could withstand high temperature operating conditions over 700°C. This study reported damaging defects for larger dimensions. Pelanconi et al. [108] developed tubular heat exchangers with alumina and lattice structure using three different cellular architectures. The authors applied SLA technique to fabricate the lattices and achieved 160% to 280% optimized heat transfer between cold flowing air and the ceramic tube with close agreement between CFD and experimental results [108]. Despite numerous challenges, it is expected that powder-based AM will become the dominant technology to fabricate ceramics in near future [104]. This could be very beneficial for HX community.

3.1.3. Polymers

Polymers by far are the most used AM material. Favorable properties such as low cost, relatively low weight, anti-fouling and anticorrosion has distinguished the application of polymers. However, low thermal conductivity is the major concern for widespread application of polymers in HX systems. To address this issue, Cevallos [35] introduced thermally enhanced polymers. Arie et al. [41] fabricated an air-to-water polymer heat exchanger by polyethylene sheets. Experimental tests in [41] demonstrated that polymer HX could provide equal or superior performance for the same pressure drop compared to plain plate fin surfaces. They measured an overall heat transfer coefficient of 35-120 W/m²K for an air-side flow rate of 3-24 l/s and a water-side flow rate of 12.5 m l/s. This research group has also worked on a cross media approach and developed a polymer composite HX with over 200% improvement in heat flow rate over mass [109]. Other than polymer

heat exchangers, composite heat exchangers additively manufactured from a polymer and a high conductive metal fibers is also reported in literature [110]. A comprehensive review of polymeric HXs are presented in [111, 112]. Here, the challenges and opportunities are highlighted for the sake of comparison.

3.2 Cold Spray

This section provides more details about CS as a promising technique to additively manufacture HXs. The CS process uses low-temperature high-pressure gas jet to accelerate particles with micrometer scale size. The low processing temperature is helpful to (i) produce oxide-free deposits, (ii) retain the feedstock microstructures and properties, (iii) diminish adverse effect on the underlying build surface, (iv) omit the need for a protected environment [20, 113]. The metallurgical bonding is merely caused by kinetic energy with high strain rate plastic deformation and sporadic powder melting. Therefore, ideal materials for CS applications have low melting point and low mechanical strength (e.g. Al, Zn, Cu). This is a limitation for CS as the majority of high strength materials such as Ti and Ni need sufficient energy for successful bonding.

CS has found fabrication and repair applications in AM. It has a great potential to be used for HX applications in order to reduce cost, provide higher production rate, and enable large-scale implementation [86]. To fabricate pin fins array, a mask is applied to selectively shield the build surface [47], as shown in Figure 4. Using this techniques, tapered pin fins such as pyramids or cones can be fabricated [47, 53, 54]. Cormier et al. [96] applied cold spray to produce square base pyramidal pin fin arrays. They concluded that higher fins would increase thermal conductance at low Reynolds numbers, but the pressure loss needs further investigations. The challenges and considerations in cold spray AM are quite similar to other AM technologies and include

- Need for post-machining process and post-treatment methods [113], which was not extensively addressed in the literature.
- A thorough understanding of anisotropic properties and residual stress emanated by high velocity impact. In this respect, Yang et al. [114] experimentally showed the existence of mechanical anisotropy in cold-sprayed copper bulk.
- A systematic investigation of the spraying parameters (e.g. scanning steps and gas parameters) to understand and optimize their influence on the deposit properties.

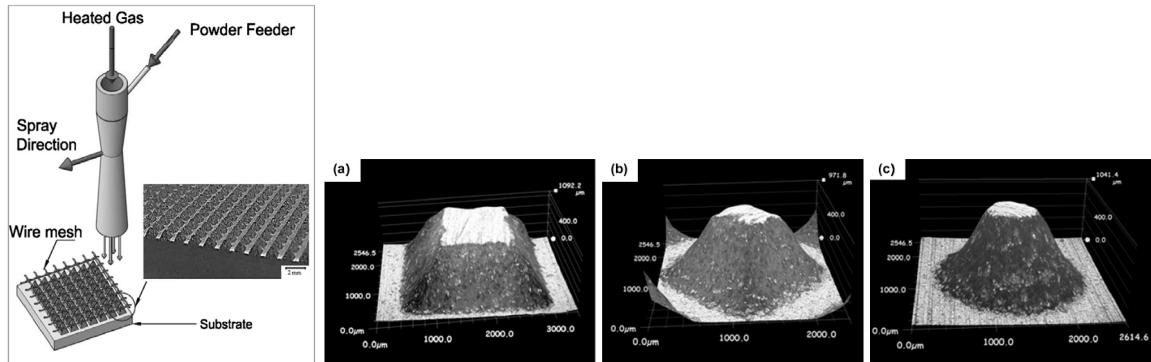


Figure 4. Pyramidal pin fin (a) square base, (b) diamond base, (c) round base fabricated in masking spray process [47]

3.3 Fabrication opportunities and challenges

AM technologies have many advantages which can be directly applied to HX design and manufacturing. First and foremost, AM has the potential to fabricate highly customized freeform features. It also allows multi-material fabrication and design optimization with minimal added manufacturing costs. Depending on the design and machine capability, AM technologies offer faster design process and higher production rating along with saving in material and energy.

Nonetheless, there are many design-related considerations that affect the manufacturability and need to be taken into account. For example, thicker walls are required for low strength materials; the orientation of some features might change due to limited build volume; horizontal channels may face size limit; and fins may experience crack or thermal distortion. Furthermore, the size and topology of certain features would impact other design features in metal AM. For instance, spacing in HX channels is the determining factor for pins' shapes [10]. The interested readers are referred to the existing design guidelines [115, 116]. This section confers the opportunities and challenges that HX community is facing to fully benefit from metal AM.

Surface roughness characterization, particularly for micro geometries, is still a major challenge in metal AM. Surface roughness is comparatively high in various metal AM technologies [53, 117]. In effect, multiple factors contribute to surface finish of AM components [10, 59, 117]. Process parameters such as energy density, layer thickness, and scanning speed, are the main sources of controlling surface roughness [118, 119]. There are complicated non-linear interactions between process parameters in laser sintering [120]. Any process parameter with impact on the partial melting of the powders can affect the surface roughness [59]. To name a few, build direction [121, 122], material [122], size of powder particles [59], arrangement of parts in the buildup chamber and heat accumulation [59], sloping angle [117], types of features (i.e. downward vs. upward-facing) [123], and support for overhanging features are among the factors influencing surface roughness. Rough surface finish was reported for downward facing surfaces due to higher heat accumulation [10]. Accordingly, process parameters such as laser power are more significant in controlling the surface quality of downward facing surfaces and features in open channel space [123]. Mumtaz and Hopkinson investigated multilayer thin-wall Inconel 625 parts and discovered that parameter changes to decrease the top surface roughness might increase the side wall roughness [124]. Moreover, it should be noted that unmelted powders cause higher surface roughness and crack initiation [125].

Experimental results reported in recent publications disclose the strong influence of surface roughness on HX efficiency [10, 21, 90]. Rough surfaces proved to provide higher heat transfer and higher pressure losses over a large range of Reynolds numbers [126, 127]. In practice, higher surface roughness may cause disruption of thermal boundary layer and early transition to turbulence [10, 92]. Ventola et al. produced artificial surface roughness in heat sink fins through SLM which resulted in 35% average increase in heat transfer [92]. The authors in [92] observed on average 80% better convective heat transfer enhancement on rough flat surfaces compared to rough finned surfaces. A detailed discussion on surface roughness characterization is presented in Section 4.

The main challenge in metal AM is the near-net-shape fabrication. One can always expect a geometric difference between a printed object and the CAD known as stair-stepping effect [59]. This demands an essential post-processing which might be more expensive and more time-

consuming than the printing process. In practice, the post processing is more challenging for inclined surfaces, internal surfaces of microchannels, pin fin arrays, thin cross-sectional areas, and support removal. Moreover, the necessity of support structures add complexity to the design process. It can also increase the total cycle time and the total fabrication cost. In effect, there are features which are left unsupported. Those features are difficult to access and will end up with rough surfaces. Some features, e.g. downward facing surfaces, can be oriented at a 45° angle to ensure partial support. Researchers have been working on overhang angles to reduce the need for support mainly as a two-dimensional problem [65, 128]. More research is needed in this regard to ensure manufacturability for three-dimensional fabrication [66]. To this end, an appropriate setting of a variety of process parameters is still a major challenge toward desirable AM product. The parameters include but not limited to powder characteristics, heat power, beam diameter, scanning speed, layer thickness, and hatch spacing. In practice, these parameters have direct effect on the build time as well as the duration of exposure to the heat source which has impact on microstructural properties and dimensional variances.

Another important fabrication challenge is the cleaning of complex internal channels. For selective laser melting (SLM), powder entrapped or sintered in the channels and must be cleaned before further heat treatment or processing. If the channels have no opening or outlet, powder removal will be a challenge. For direct energy deposition (DED), the low heat flux and possibility to adjust the optics can control the distortion and minimize the problem of powder removal in the future.

4. Experimental evaluation of Additively Manufactured HXs

Although AM offers new design opportunities, it is still essential to characterize the HXs performance through experiments or in actual applications. Experimental characterization of thermal and hydrodynamic performance of HXs has been investigated by a handful of researchers, for instance [1], [3], [10], [47]. This section is about to highlight the significant insights in experimental evaluation of additively manufactured HXs. The important considerations in such experimental evaluations include, but are not limited to, characterizing flow structures and flow maldistribution, measuring pressure drop inside a manifold, determining air side pressure drop, scaling up microchannel heat exchangers, producing velocity profiles, determining sources of experimental uncertainty, testing over large range of Reynolds numbers, using conventional HX correlations for newly designed AM heat exchangers, comparing baseline and AM heat exchangers in various flow rates, flow structure visualization, analyzing Nusselt number and friction factor, characterizing the turbulence intensity and the turbulent kinetic energy for miniature geometries, e.g. in fin array [47], and analyzing the impact of turbulence quantities on thermal and hydrodynamic performance in compact HX design [47].

As discussed in earlier sections, AM leaves rough surfaces. Such surface roughness can affect the hydrodynamic and heat transfer performance of AM parts. Many studies have reported roughness-induced early transition from laminar flow to turbulent flow [47, 53, 90, 93, 96, 129, 130]. Stimpson et al. [90] measured air pressure drop through AM channels with design hydraulic diameters between 635 and 1261 μm . Based on air flow pressure drop, they reported significant augmentation of friction factor in additively manufactured coupons compared to smooth channels.

Figure 5 shows variation of the friction factor, f , based on the Reynolds number reported in Ref. [90]. The friction factor was calculated based on Equation 1.

$$f = \Delta P \frac{2D_h}{L\rho V^2} \quad (1)$$

where D_h , ρ , L and V are the hydraulic diameter, density, length of pressure drop measurement, and the velocity, respectively.

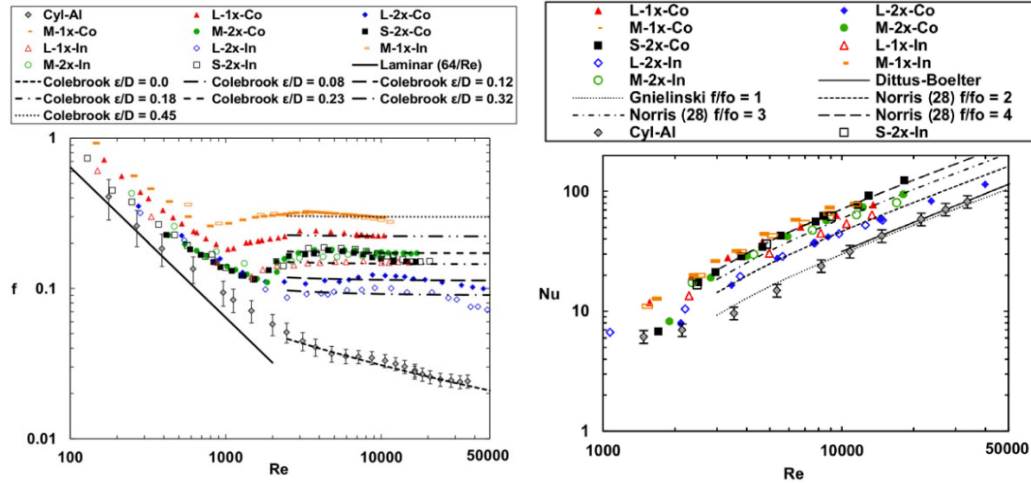


Figure 5. (Left) Friction factor, (Right) Nusselt number for different coupons used in [90]

Figure 5 compares the experimentally obtained friction factor to the laminar flow theory friction factor. Results suggest a significant augmentation in friction factor which is caused by roughness features in AM flow channels. For laminar flow regime with the Reynolds number less than around 2000, the friction factor based on the laminar flow theory and for smooth surfaces ($f=64/Re$) is shown with the solid line. The figure shows the experimentally obtained friction factors for AM coupons are greater than the theoretical friction factor for smooth surfaces. The plot on right demonstrates the heat transfer performance of AM coupons by comparing the Nusselt numbers (Nu) from AM coupons to the Dittus-Boelter correlation, Equation 2, as well as the Gnielinski correlation, Equation 3.

$$Nu = 0.023 Re^{0.8} Pr^{0.4} \quad (2)$$

$$Nu = \frac{\frac{f}{8}(Re-1000)Pr}{1+12.7\sqrt{\frac{f}{8}}(Pr^{\frac{2}{3}}-1)} \quad (2)$$

This plot also shows that the order of the data from the lowest Nu to the highest Nu match with the order of friction factor data. Stimpson et al. [131] concluded that while the pressure loss values (and friction factor) increase with Reynolds number little or no increase occurs in heat transfer performance. The same result is suggested in many other studies as listed in Table 2. This table summarizes literature on hydrodynamic and heat transfer performance of AM heat exchangers and flow channels.

Table 2: Review of the literature evaluating hydrodynamic and/or heat transfer performance of AM heat exchangers and flow channels

Ref.	AM Method	Coupon Features	Experimental Study	Significant Observations
[90]	DMLS	• Coupons were designed with flow channels inside • Materials were Al, CoCr, and Inconel 718 • Hydraulic diameter between 635 and 1261 μm	• Characterizing surface roughness by measuring pressure drop • Evaluating heat transfer performance • The heat transfer performance was also studied by comparing the Nusselt number to the Dittus-Boelter correlation and also the Gnielinski correlation.	• Significant augmentation in friction factor in AM flow channels • Friction factors were greater than the laminar flow theory ($64/\text{Re}$) and the Colebrook equation for the laminar and turbulent flow, respectively. • The obtained Nusselt numbers were within $\pm 12\%$ of both correlations. • Augmentation of Nusselt number (heat transfer) did not increase proportionally with the augmentation of the friction factor.
[98]	Laser-based powder bed fusion	• Coupons were designed with flow channels inside • Coupons had film cooling holes	• The discharge coefficient through holes were obtained by supplying the coolant fluid • Comparison was done on discharge coefficient between AM coupons and EDM-fabricated coupons	• The discharge coefficients of film cooling holes obtained from EDM were greater than that in AM coupons. This was attributed to a higher surface roughness in AM flow channels. • It was observed that while the pressure loss values increase with Reynolds number little or no increase occurs in heat transfer performance.
[93, 130]	SLM	• Straight and manifold microchannel heat sinks with hydraulic diameters of 500 μm • Water was supplied with mass fluxes between 500 and 2000 $\text{kg}/\text{m}^2\text{s}$	• Water was supplied with mass fluxes between 500 and 2000 $\text{kg}/\text{m}^2\text{s}$ • Water pressure drop along the channel was measured. • The Fanning friction factor was obtained from the pressure drop. • Surface roughness was measured based on interferometry measurements.	• An early transition from the laminar flow to the turbulent flow at a critical Reynolds number around 600 was observed. • This early transition to the turbulent flow was attributed to the roughness features in AM flow channels. • An exterior surface roughness around 20 μm was reported. • Higher surface roughness values were reported for interior surfaces due to “burn” which was defined by partial sintering of loose powder to nearby solid features that can be exacerbated by heat build-up within the part during fabrication.
[132]	laser-based powder bed fusion	• Heat exchangers were fabricated from AlSi10Mg. • Plate-fin air–liquid crossflow heat exchangers which were originally used for the aircraft oil cooler application were additively manufactured.	• X-ray computed tomography scans were performed to replicate internal features of the heat exchanger. • Air-side pressure drop was measured	• Higher pressure drop values were measured in AM heat exchangers compared to the one that was manufactured based on conventional manufacturing methods. • The higher-pressure drop was attributed to the roughness features in flow channels.

[133]	DMLS	• A novel permeable membrane microchannel (PMM) heat sink was designed and fabricated by AM.	• The hydraulic performance of the heat sink was evaluated.	• The permeable design lowered the pressure drop between 20% and 70% compared to a traditional manifold microchannel heat exchanger design.
[131]	DMLS	• Coupons were designed with flow channels inside • Hydraulic diameter between 635 and 1261 μm	• CT scanning, scanning white light interferometer, and scanning electron microscopy (SEM) was used to examine surface roughness values of the coupons.	• Larger surface roughness values were reported for downward facing coupons compared to upward-facing and vertical coupons.
[121]	SLM	• Test coupons were fabricated in different orientation of vertical, horizontal, and diagonal. • Coupons had circular channels with diameters around 500 μm.	• Effects of build direction on surface roughness of circular microchannels with diameters around 500 μm were studied. • X-ray CT scanner was used to inspect the interior surfaces of channels. • Effects of build direction on channels with different cross-sectional geometries of (i) cylindrical, (ii) diamond, and (iii) teardrop were evaluated. • The heat transfer performance was also evaluated for coupons.	• Channels fabricated in vertical orientation showed lowest surface roughness features. • Channels fabricated in horizontal and diagonal orientations showed almost identical roughness features. • A roughness-induced early transition from laminar flow to turbulent flow was reported. • For turbulent flow regime and for channels manufactured horizontally the highest and the lowest friction factor values belonged to cylindrical and teardrop geometries, respectively. • The heat transfer performance was similar between the three different build directions. • The channel geometry had negligible impact on heat transfer performance. • Augmentation of Nusselt number did not increase proportionally with the augmentation of the friction factor.
[91]	DMLS	• Coupons with pin fins in triangular, star, and dimpled spheres were manufactured in pin fin arrays	• Evaluated effects of geometry, spacing, and number of pin fins in pin fin arrays • CT scanner was used to obtain images of the interior of the coupons • Pressure drop across the arrays were also measured	• The order of pressure drop values from the lowest to highest were (i) point-oriented triangle pin fins, (2) star-shaped pin fins, and (iii) face-oriented triangle pin fins. • An effect of streamwise spacing on the hydrodynamic performance of the coupons was observed and attributed to the size of the wake generated by the pin.
[10]	Laser powder bed fusion	• Pin fin arrays with four different spacings were manufactured. • Pin fins were designed to be cylindrical in shape with a diameter of 1 mm and a height to diameter ratio of 1.	• pressure loss and heat transfer measurements were evaluated over a range of Reynolds numbers. • The test coupons were evaluated nondestructively and were compared to the original CAD model.	• The friction factor values were around 15 times the friction factor obtained from the Colebrook equation for the turbulent flow. • The heat transfer increased in arrays with tighter span-wise spacing. • Higher heat transfer was reported for AM pin fins compared to smooth channel benchmark. This was attributed to higher surface roughness which was more significant at higher Reynolds numbers.

				Span-wise spacing was observed to strongly affect the heat transfer due to its impact on wake formation and interaction among the pins.
[98]	DMLS	- Coupons were made from a high temperature nickel alloy. - Coupons were made at two different hole sizes (internal hydraulic diameters of 1.26 mm and 2.45 mm) and two different build directions (vertical and angled).	- They studied roughness of film cooling holes. - SEM micrograph of holes were obtained and compared with a hole manufactured by EDM as the benchmark	- The EDM hole had smooth surfaces compared to AM holes. - The holes fabricated by AM demonstrated large roughness features which could block the flow.
[51]	SLM	Fabricated five different heat sink pin arrays; (1) cylindrical pin fins, (2) rectangular pin fins, (3) staggered pin fins, (4) lattice pin fins, and (5) rectangular pin fins with rounded corners.	The hydrodynamic and heat transfer performance of the heat sinks were evaluated experimentally.	Results indicated a superior heat transfer performance for the rectangular pin fin with rounded corners. The pressure drop for this design was lower than the rectangular pin fin arrays.
[134]	Extrusion	The flow channels were made of 150 μm-thick high-density polyethylene sheets. Channels had a width of 15.5 cm and were 29 cm long.	An air-water polymer heat exchanger was made of thin polymer sheets using layer-by-layer line welding with a laser	- An overall heat transfer coefficient of 35-120 W/m²K was achieved for an air flow rate between 3 and 24 L/s and a water flow rate of 12.5 mL/s. - Because of the thin walls in the heat exchanger (150 μm), the wall thermal resistance was calculated to account for only 3% of the total thermal resistance.
[135]	DMLS	- A large prototype of a manifold-microchannel heat exchanger was additively manufactured. - The heat exchanger was manufactured from Ti64 and had an overall dimension of 15 x 15 x 3.2 cm.	Heat transfer performance of the heat exchanger was evaluated.	Improvements up to 80%, 120%, and 190% were reported for air-side heat transfer coefficient when compared with wavy fin, louvered fin, and plain plate fin heat exchangers, respectively.
[28, 110, 136]	DMLS	- Three manifold-microchannel heat exchangers were additively manufactured. - Coupons were fabricated from Ti64, SS17-4, and AlSi10Mg. - Microchannels had a thickness of 150 μm and were tilted 45° to avoid unsupported overhangs. The microchannel width was 0.27 mm.	The hydrodynamic and heat transfer performance of three heat exchangers were evaluated experimentally.	- For the range of air flow rates tested, heat exchanger capacities of up to 200, 300, and 400 W were obtained for the Ti64, SS17-4, and AlSi10Mg heat exchangers, respectively. - A lower air-side pressure drop was observed for Ti64 heat exchanger compared to the other two heat exchangers. This was attributed to smoother surfaces on the Ti64 coupon.
[88]	SLM	- A manifold-microchannel was manufactured by AM. - The HX was manufactured from Inconel 718 with a core	The HX was tested with nitrogen gas at 600°C on the hot side and air at 38°C on the cold side.	A maximum heat duty of 2.78 kW and a maximum overall heat transfer coefficient of 1000 W/m²K with a COP of 62 was achieved.

		size of 66 x 74 x 27 mm. The minimum fin thickness was 180 μm .		• Up to 25% increase in heat transfer density was achieved in AM heat exchanger compared to plate fin heat exchanger.
[3]	Extrusion	• An AM manifold, fabricated from ABS, was mounted around a commercially available enhanced tube with fin structure on its outer surface and rifling structure on its inner bore.	• The HX was tested with water in both shell and tube side.	• An overall heat transfer coefficient of 22,000 W/m²K was achieved.

5. Summary and conclusions

An ideal HX design ensures balance among design specifications (i.e. size, weight, material), performance characteristics (i.e. heat transfer and pressure drop), and other requirements such as reliability and maintainability. The primary objective in HX design is enhancing heat transfer rate at a minimal pressure drop (improving energy efficiency and power density). AM offers tremendous advantages to the design and manufacturing of HXs including

- An ability to fabricate optimized geometries with enhanced surfaces. In essence, AM has provided the opportunity of exploring novel designs and overcoming the limitations of conventional fabrication methods.
- Consolidation and monolithic bonds which eliminates the need for weld or braze and may reduce the production cycle time. In essence, inappropriate welding or brazing may result in clogging (e.g. in microchannels). It is helpful in improving reliability particularly in high pressure applications. Furthermore, integrating HPs into the structure would be beneficial for novel energy storage systems.
- The opportunity for controlled surface roughness with the goal of heat transfer enhancement through innovative heat transfer surface designs. Designers of additive manufacturing parts must realize how additive manufacturing can affect surface roughness as it impacts the amount of cooling that can be achieved in actual applications. The existing research proves that upward-facing surfaces have smaller roughness than downward-facing surfaces. Further development is needed to explore the desirable surface roughness in the design phase and characterize the effects of increased surface roughness on thermal efficiency. In this respect, particular attention should be paid to adjoining internal surfaces of microchannels, e.g. pin fin arrays, which are not easily accessible.
- Unique opportunity for fully controlled and organized porous structures. Porous structures (including lattice structures) are favorable for conductive heat transfer improvement.

Despite various research efforts into applying AM technologies for HXs, it is clear that there are still many research opportunities that require further investigation:

- Materials are still the cornerstone of AM development and needs more fundamental research particularly for high temperature HX applications. In this respect, the literature

suggests investigating more aerospace-grade materials, ceramic material with two-dimensional reinforcement, copper, certain Haynes alloys, and even composites. It is recognized that low temperature conventional heat exchanger designs are uneconomical in high temperature ($>500^{\circ}\text{C}$) applications as they encounter creep, reduced strength, oxidation, and thermal shock.

- Traditional heat exchangers use the configuration of shell and tube, plate and fin, or plate and frame. AM offers unprecedented design freedom to fabricate HX with customized geometries. However, the so-called assertion of “complexity is free in AM” would not work well when it comes to internal features. The degree of complexity for internal feature is dependent on manufacturing process capability.
- Design for AM requires a particular thought process as well as thorough understanding of various AM technologies. The literature suggests innovative design-for-additive solutions for fabrication beneficial features and achieving desirable surface conditions especially for mini- and micro-channels. It is advantageous to combine topology optimization with AM methods for metal with higher thermal conductivity [51]. Design and manufacturing of complex and compact HX through AM can be easily justified if it can be extended to new and wide range of applications.
- Despite tremendous progress, fabrication inaccuracy and adequate reproducibility will remain a key issue for micro geometries.
- AM technologies and applications have experienced continuous advancements in the past decade. In near future, one can expect enhanced manufacturability, sophisticated simulation packages, and advanced process control tools for metal AM. In particular, accurate simulation is needed to quantify heat performance in multiphase HXs with porous media, capillary structures, and manifold-microchannel HXs.
- Thus far, SLM has been the most popular metal AM technology for HX applications mainly because of high-resolution and high flexibility. It seems SLM will keep its popular position particularly for innovative passive thermal systems. Nevertheless, enhanced SLM systems with less production time and cost would be very advantages for large scale production of additively manufactured HXs.
- Performance characterization will remain a challenge for a newly developed HXs particularly for complex two-phase devices. In this respect, more data is needed to evaluate the thermal performance of additively made HXs. In addition, it should be noted that in the new HX design trend minimizing pressure drop has not been considered as much as increasing the surface to volume ratio.
- New methods of reliability analysis are required for miniature geometries and thin walls which are being used in novel compact HX designs.

Acknowledgment

We would like to thank our colleagues at Connecticut Center for Advanced Technology and Advanced Digital Design and Fabrication Lab at the University of Massachusetts Amherst for their professional insights and suggestions. Dongsheng Li's contribution is based upon work supported

by the U.S. Department of Energy's Office of Energy Efficiency and Renewable Energy (EERE) under the Fuel Cell Technologies Office Award DE-0008423.

References

1. Saltzman DJ, Bichnevicius M, Lynch SP, Simpson T, Reutzel T, Dickman C, Martukanitz R (2017) Experimental comparison of a traditionally built versus additively manufactured aircraft heat exchanger. 55th AIAA Aerosp Sci Meet. doi: 10.2514/6.2017-0902
2. Romei F, Grubišić AN, Gibbon D (2017) Manufacturing of a high-temperature resistojet heat exchanger by selective laser melting. *Acta Astronaut* 138:356–368
3. Tiwari R, Andhare RS, Shooshtari A, Ohadi M (2019) Development of an Additive Manufacturing-Enabled Compact Manifold Microchannel Heat Exchanger. *Appl Therm Eng* 147:781–788
4. Zhang X, Keramati H, Arie M, Singer F, Tiwari R, Shooshtari A, Ohadi M (2018) Recent developments in high temperature heat exchangers: A review. *Front Heat Mass Transf* 11:18
5. Jafari D, Wits WW (2018) The utilization of selective laser melting technology on heat transfer devices for thermal energy conversion applications: A review. *Renew Sustain Energy Rev* 91:420–442
6. Moon H, Miljkovic N, King WP (2020) High power density thermal energy storage using additively manufactured heat exchangers and phase change material. *Int J Heat Mass Transf* 153:119591
7. Dede EM, Joshi SN, Zhou F (2015) Topology Optimization, Additive Layer Manufacturing, and Experimental Testing of an Air-Cooled Heat Sink. *J Mech Des* 137:111403
8. Li D, Maloney T, Mannan N, Niknam S (2020) Design of additively manufactured methanol conversion reactor for high throughput production. *Mater Des Process Commun*. doi: 10.1002/mdp2.143
9. Sabau AS, Bejan A, Brownell D, Gluesenkamp K, Murphy B, List F, Carver K, Schaich CR, Klett JW (2020) Design, additive manufacturing, and performance of heat exchanger with a novel flow-path architecture. *Appl Therm Eng* 180:115775
10. Kirsch KL, Thole KA (2017) Pressure loss and heat transfer performance for additively and conventionally manufactured pin fin arrays. *Int J Heat Mass Transf* 108:2502–2513
11. Chan CW, Siqueiros E, Ling-Chin J, Royapoor M, Roskilly AP (2015) Heat utilisation technologies: A critical review of heat pipes. *Renew Sustain Energy Rev* 50:615–627
12. ISO/ASTM (2015) ISO/ASTM 52900: Additive manufacturing - General principles - Terminology.
13. Herzog D, Seyda V, Wycisk E, Emmelmann C (2016) Additive manufacturing of metals. *Acta Mater* 117:371–392
14. DebRoy T, Wei HLL, Zuback JSS, Mukherjee T, Elmer JWW, Milewski JOO, Beese

- AMM, Wilson-Heid A, De A, Zhang W (2018) Additive manufacturing of metallic components – Process, structure and properties. *Prog Mater Sci* 92:112–224
15. 52921 I-A (2013) Standard Terminology for Additive Manufacturing-Coordinate Systems and Test Methodologies.
 16. Gradl PR (2016) Rapid Fabrication Techniques for Liquid Rocket Channel Wall Nozzles. 52nd AIAA/SAE/ASEE Jt Propuls Conf. doi: 10.2514/6.2016-4771
 17. Gradl PR, Greene SE, Protz C, Bullard B, Buzzell J, Garcia C, Wood J, Cooper K, Hulka J, Osborne R (2018) Additive manufacturing of liquid rocket engine combustion devices: A summary of process developments and hot-fire testing results. 2018 Jt Propuls Conf. doi: 10.2514/6.2018-4625
 18. Gradl PR, Protz C, Wammen T (2019) Additive Manufacturing Development and Hot-fire Testing of Liquid Rocket Channel Wall Nozzles using Blown Powder Directed Energy Deposition Inconel 625 and JBK-75 Alloys. 55th AIAA/SAE/ASEE Jt. Propuls. Conf.
 19. Gradl PR, Greene SE, Wammen T (2019) Bimetallic channel wall nozzle development and hot-fire testing using additively manufactured laser wire direct closeout technology. AIAA Propuls Energy Forum Expo 2019. doi: 10.2514/6.2019-4361
 20. AlMangour B (2018) Fundamentals of Cold Spray Processing: Evolution and Future Perspectives. In: *Cold-Spray Coatings*. Springer International Publishing, Cham, pp 3–24
 21. Kirsch KL, Thole KA (2018) Numerical Optimization, Characterization, and Experimental Investigation of Additively Manufactured Communicating Microchannels. *J Turbomach* 40:111003
 22. Esarte J, Blanco JM, Bernardini A, San-José JT (2017) Optimizing the design of a two-phase cooling system loop heat pipe: Wick manufacturing with the 3D selective laser melting printing technique and prototype testing. *Appl Therm Eng* 111:407–419
 23. Ameli M, Agnew B, Leung PS, Ng B, Sutcliffe CJ, Singh J, McGlen R (2013) A novel method for manufacturing sintered aluminium heat pipes (SAHP). *Appl Therm Eng* 52:498–504
 24. Ho JY, Wong KK, Leong KC (2016) Saturated pool boiling of FC-72 from enhanced surfaces produced by Selective Laser Melting. *Int J Heat Mass Transf* 99:107–121
 25. Wong KK, Leong KC (2018) Saturated pool boiling enhancement using porous lattice structures produced by Selective Laser Melting. *Int J Heat Mass Transf* 121:46–63
 26. Wong KK, Leong KC (2019) Nucleate flow boiling enhancement on engineered three-dimensional porous metallic structures in FC-72. *Appl Therm Eng* 159:113846
 27. Neugebauer R, Müller B, Gebauer M, Töppel T (2011) Additive manufacturing boosts efficiency of heat transfer components. *Assem Autom* 31:344–347
 28. Arie MA, Shooshtari AH, Rao V V., Dessiatoun S V., Ohadi MM (2016) Air-Side Heat Transfer Enhancement Utilizing Design Optimization and an Additive Manufacturing Technique. *J Heat Transfer* 139:031901

29. Yameen WC, Piascik NA, Clemente RC, Miller AK, Niknam SA, Benner J, Santamaria AD, Mortazavi M (2019) Experimental Characterization of a Manifold-Microchannel Heat Exchanger Fabricated Based on Additive Manufacturing. In: 2019 18th IEEE Intersoc. Conf. Therm. Thermomechanical Phenom. Electron. Syst. IEEE, pp 621–625
30. Guo X, Fan Y, Luo L (2014) Multi-channel heat exchanger-reactor using arborescent distributors: A characterization study of fluid distribution, heat exchange performance and exothermic reaction. *Energy* 69:728–741
31. Fasano M, Ventola L, Calignano F, Manfredi D, Ambrosio EP, Chiavazzo E, Asinari P (2016) Passive heat transfer enhancement by 3D printed Pitot tube based heat sink. *Int Commun Heat Mass Transf* 74:36–39
32. Ho JY, Wong KK, Leong KC, Wong TN (2017) Convective heat transfer performance of airfoil heat sinks fabricated by selective laser melting. *Int J Therm Sci* 114:213–228
33. Hathaway BJ, Garde K, Mantell SC, Davidson JH (2018) Design and characterization of an additive manufactured hydraulic oil cooler. *Int J Heat Mass Transf* 117:188–200
34. Bacellar D, Aute V, Huang Z, Radermacher R (2017) Design optimization and validation of high-performance heat exchangers using approximation assisted optimization and additive manufacturing. *Sci Technol Built Environ* 23:896–911
35. Cevallos JG (2014) Thermal and Manufacturing Design of Polymer Composite Heat Exchangers. University of Maryland
36. Cárdenas B, Garvey S, Kantharaj B, Simpson M (2017) Parametric investigation of a non-constant cross sectional area air to air heat exchanger. *Appl Therm Eng* 113:278–289
37. Syed-Khaja A, Freire AP, Kaestle C, Franke J (2017) Feasibility Investigations on Selective Laser Melting for the Development of Microchannel Cooling in Power Electronics. In: 2017 IEEE 67th Electron. Components Technol. Conf. IEEE, pp 1491–1496
38. Gerstler WD, Erno D (2017) Introduction of an additively manufactured multi-furcating heat exchanger. In: 2017 16th IEEE Intersoc. Conf. Therm. Thermomechanical Phenom. Electron. Syst. IEEE, pp 624–633
39. Bernardin JD, Ferguson K, Sattler D, Kim S-J (2017) The Design, Analysis, and Fabrication of an Additively Manufactured Twisted Tube Heat Exchanger. ASME 2017 Heat Transf. Summer Conf.
40. Ibrahim OT, Monroe JG, Thompson SM, Shamsaei N, Bilheux H, Elwany A, Bian L (2017) An investigation of a multi-layered oscillating heat pipe additively manufactured from Ti-6Al-4V powder. *Int J Heat Mass Transf* 108:1036–1047
41. Arie MA, Shooshtari AH, Tiwari R, Dessiatoun S V., Ohadi MM, Pearce JM (2017) Experimental characterization of heat transfer in an additively manufactured polymer heat exchanger. *Appl Therm Eng* 113:575–584
42. Gutmann B, Köckinger M, Glotz G, Ciaglia T, Slama E, Zadavec M, Pfanner S, Maier MC, Gruber-Wölfler H, Oliver Kappe C (2017) Design and 3D printing of a stainless steel

- reactor for continuous difluoromethylations using fluoroform. *React Chem Eng* 2:919–927
43. Aris MS, Owen I, Sutcliffe CJ (2011) The development of active vortex generators from shape memory alloys for the convective cooling of heated surfaces. *Int J Heat Mass Transf* 54:3566–3574
 44. Kirsch KL, Thole KA (2017) Experimental Investigation of Numerically Optimized Wavy Microchannels Created Through Additive Manufacturing. *J Turbomach* 140:021002
 45. Kirsch KL, Thole KA (2016) Heat Transfer and Pressure Loss Measurements in Additively Manufactured Wavy Microchannels. *J Turbomach* 139:011007
 46. Bergman TL, Incropera FP (2011) Fundamentals of heat and mass transfer. Wiley
 47. Dupuis P, Cormier Y, Fenech M, Jodoin B (2016) Heat transfer and flow structure characterization for pin fins produced by cold spray additive manufacturing. *Int J Heat Mass Transf* 98:650–661
 48. Sahiti N, Durst F, Dewan A (2005) Heat transfer enhancement by pin elements. *Int J Heat Mass Transf* 48:4738–4747
 49. Şara O. (2003) Performance analysis of rectangular ducts with staggered square pin fins. *Energy Convers Manag* 44:1787–1803
 50. Alsulami M, Mortazavi M, Niknam SA, Li D (2020) Design complexity and performance analysis in additively manufactured heat exchangers. *Int J Adv Manuf Technol* 110:865–873
 51. Wong M, Owen I, Sutcliffe CJ, Puri A (2009) Convective heat transfer and pressure losses across novel heat sinks fabricated by Selective Laser Melting. *Int J Heat Mass Transf* 52:281–288
 52. Robinson AJ, Tan W, Kempers R, Colenbrander J, Bushnell N, Chen R (2017) An ultra high performance heat sink using a novel hybrid impinging microjet — Microchannel structure. In: 2017 16th IEEE Intersoc. Conf. Therm. Thermomechanical Phenom. Electron. Syst. IEEE, pp 482–490
 53. Dupuis P, Cormier Y, Fenech M, Corbeil A, Jodoin B (2016) Flow structure identification and analysis in fin arrays produced by cold spray additive manufacturing. *Int J Heat Mass Transf* 93:301–313
 54. Dupuis P, Cormier Y, Farjam A, Jodoin B, Corbeil A (2014) Performance evaluation of near-net pyramidal shaped fin arrays manufactured by cold spray. *Int J Heat Mass Transf* 69:34–43
 55. Wang XW, Ho JY, Leong KC (2018) An experimental investigation of single droplet impact cooling on hot enhanced surfaces fabricated by selective laser melting. *Int J Heat Mass Transf* 120:652–670
 56. Thompson SM, Aspin ZS, Shamsaei N, Elwany A, Bian L (2015) Additive manufacturing of heat exchangers: A case study on a multi-layered Ti–6Al–4V oscillating heat pipe. *Addit Manuf* 8:163–174

57. Jafari D, Wits WW, Geurts BJ (2017) An investigation of porous structure characteristics of heat pipes made by additive manufacturing. In: 2017 23rd Int. Work. Therm. Investig. ICs Syst. IEEE, pp 1–7
58. Wang D, Yang Y, Liu R, Xiao D, Sun J (2013) Study on the designing rules and processability of porous structure based on selective laser melting (SLM). *J Mater Process Technol* 213:1734–1742
59. Jamshidinia M, Kovacevic R (2015) The influence of heat accumulation on the surface roughness in powder-bed additive manufacturing. *Surf Topogr Metrol Prop* 3:014003
60. Sing SL, Wiria FE, Yeong WY (2018) Selective laser melting of lattice structures: A statistical approach to manufacturability and mechanical behavior. *Robot Comput Integr Manuf* 49:170–180
61. Wauthle R, Vrancken B, Beynaerts B, Jorissen K, Schrooten J, Kruth J-P, Van Humbeeck J (2015) Effects of build orientation and heat treatment on the microstructure and mechanical properties of selective laser melted Ti6Al4V lattice structures. *Addit Manuf* 5:77–84
62. Ho JY, Leong KC, Wong TN (2019) Experimental and numerical investigation of forced convection heat transfer in porous lattice structures produced by selective laser melting. *Int J Therm Sci* 137:276–287
63. Cheng L, Liu J, Liang X, To AC (2018) Coupling lattice structure topology optimization with design-dependent feature evolution for additive manufactured heat conduction design. *Comput Methods Appl Mech Eng* 332:408–439
64. Liang X, Cheng L, Kang Z, et al (2018) Current and future trends in topology optimization for additive manufacturing. *Struct Multidiscip Optim* 57:2457–2483
65. Gaynor AT, Guest JK (2016) Topology optimization considering overhang constraints: Eliminating sacrificial support material in additive manufacturing through design. *Struct Multidiscip Optim* 54:1157–1172
66. Cheng L, Bai J, To AC (2019) Functionally graded lattice structure topology optimization for the design of additive manufactured components with stress constraints. *Comput Methods Appl Mech Eng* 344:334–359
67. Koga AA, Lopes ECC, Villa Nova HF, Lima CR de, Silva ECN (2013) Development of heat sink device by using topology optimization. *Int J Heat Mass Transf* 64:759–772
68. Das S, Sutradhar A (2020) Multi-physics topology optimization of functionally graded controllable porous structures: Application to heat dissipating problems. *Mater. Des.* 193:
69. Tawk R, Ghannam B, Nemer M (2019) Topology optimization of heat and mass transfer problems in two fluids—one solid domains. *Numer Heat Transf Part B Fundam* 76:130–151
70. Kobayashi H, Yaji K, Yamasaki S, Fujita K (2020) Topology design of two-fluid heat exchange.

71. Liu J, Ma Y (2016) A survey of manufacturing oriented topology optimization methods. *Adv Eng Softw* 100:161–175
72. Alexandersen J, Andreasen CS (2020) A Review of Topology Optimisation for Fluid-Based Problems. *Fluids* 5:29
73. Atzeni E, Salmi A (2012) Economics of additive manufacturing for end-usable metal parts. *Int J Adv Manuf Technol* 62:1147–1155
74. Quinlan HE, Hasan T, Jaddou J, Hart AJ (2017) Industrial and Consumer Uses of Additive Manufacturing: A Discussion of Capabilities, Trajectories, and Challenges. *J Ind Ecol* 21:S15–S20
75. Niknam SA, Li D, Das G (2019) An acoustic emission study of anisotropy in additively manufactured Ti-6Al-4V. *Int J Adv Manuf Technol* 100:1731–1740
76. Wong M, Tsopanos S, Sutcliffe CJ, Owen I (2007) Selective laser melting of heat transfer devices. *Rapid Prototyp J* 13:291–297
77. Popovich A, Sufiiarov V, Polozov I, Borisov E, Masaylo D, Orlov A (2016) Microstructure and mechanical properties of additive manufactured copper alloy. *Mater Lett* 179:38–41
78. Zhang G, Chen C, Wang X, Wang P, Zhang X, Gan X, Zhou K (2018) Additive manufacturing of fine-structured copper alloy by selective laser melting of pre-alloyed Cu-15Ni-8Sn powder. *Int J Adv Manuf Technol* 96:4223–4230
79. Moriya S, Inoue T, Sasaki M, Nakamoto T, Kimura T, Nomura N, Kikuchi K, Kawasaki A, Kato T, Masuda I (2018) Feasibility Study on Additive Manufacturing of Liquid Rocket Combustion Chamber. *Trans JSASS Aerosp Tech Japan* 16:261–266
80. Singer F, Deisenroth DC, Hymas DM, Ohadi MM (2017) Additively manufactured copper components and composite structures for thermal management applications. In: *Proc. 16th Intersoc. Conf. Therm. Thermomechanical Phenom. Electron. Syst. ITHERM 2017*. Institute of Electrical and Electronics Engineers Inc., pp 174–183
81. Reichardt A, Dillon R, Borgonia J, Design AS-M&, 2016 U (2016) Development and characterization of Ti-6Al-4V to 304L stainless steel gradient components fabricated with laser deposition additive manufacturing. *Mater Des* 104:404–413
82. Popovich V, Borisov E, Sufiiarov V, Popovich V, Borisov E, Popovich A, Sufiiarov Vs, Masaylo D, Alzina L (2017) Functionally graded Inconel 718 processed by additive manufacturing: Crystallographic texture, anisotropy of microstructure and mechanical properties. *Mater Des* 114:441–449
83. Karnati S, Sparks TE, Liou F, Newkirk JW, Taminger KMB, Seufzer WJ LASER METAL DEPOSITION OF FUNCTIONALLY GRADIENT MATERIALS FROM ELEMENTAL COPPER AND NICKEL POWDERS.
84. Tsopanos S, Sutcliffe CJ, Owen I (2005) The Manufacture of Micro Cross-Flow Heat Exchangers by Selective Laser Melting. *Enhanc. Compact Ultra-Compact Heat Exch. Sci. Eng. Technol.*

85. Seiya W, Zhang S (2015) 3D Printing as an Alternative Manufacturing Method for the Micro-gas Turbine Heat Exchanger.
86. Assaad J, Corbeil A, Richer PF, Jodoin B (2011) Novel Stacked Wire Mesh Compact Heat Exchangers Produced Using Cold Spray. *J Therm Spray Technol* 20:1192–1200
87. Arie M, Shooshtari A, Dessiatoun S, Ohadi M (2016) Performance characterization of an additively manufactured titanium (Ti64) heat exchanger for an air-water cooling application. *Proc. ASME Heat Transf. Fluids Eng. Nanochannels, Microchannels, Minichannels Conf. HT/FE/ICNMM2016*
88. Zhang X, Tiwari R, Shooshtari AH, Ohadi MM (2018) An additively manufactured metallic manifold-microchannel heat exchanger for high temperature applications. *Appl Therm Eng* 143:899–908
89. Zhang X, Arie MA, Deisenroth DC, Shooshtari A, Dessiatoun S, Ohadi M (2015) Impact of Additive Manufacturing on Performance Enhancement of Heat Exchangers: A Case Study on An Air-to-Air Heat Exchanger for High Temperature Applications. In: IX Minsk Int. Semin. Heat Pipes, Heat Pumps, Refrig. Power Sources “, Minsk. Belarus. pp 141–153
90. Stimpson CK, Snyder JC, Thole KA, Mongillo D (2016) Roughness Effects on Flow and Heat Transfer for Additively Manufactured Channels. *J Turbomach* 138:051008
91. Ferster KK, Kirsch KL, Thole KA (2018) Effects of Geometry, Spacing, and Number of Pin Fins in Additively Manufactured Microchannel Pin Fin Arrays. *J Turbomach*. doi: 10.1115/1.4038179
92. Ventola L, Robotti F, Dialameh M, Calignano F, Manfredi D, Chiavazzo E, Asinari P (2014) Rough surfaces with enhanced heat transfer for electronics cooling by direct metal laser sintering. *Int J Heat Mass Transf* 75:58–74
93. Collins IL, Weibel JA, Pan L, Garimella S V. (2019) Evaluation of Additively Manufactured Microchannel Heat Sinks. *IEEE Trans Components, Packag Manuf Technol* 9:446–457
94. Tsopanos S, Sutcliffe CJ, Owen I (2007) The Manufacture of Micro Cross-Flow Heat Exchangers by Selective Laser Melting. *Rapid Prototyp J* 13:291–297
95. Lohan DJ, Dede EM, Allison JT (2017) Topology optimization for heat conduction using generative design algorithms. *Struct Multidiscip Optim* 55:1063–1077
96. Cormier Y, Dupuis P, Farjam A, Corbeil A, Jodoin B (2014) Additive manufacturing of pyramidal pin fins: Height and fin density effects under forced convection. *Int J Heat Mass Transf* 75:235–244
97. Perry J, Richer P, Jodoin B, Matte E (2019) Pin Fin Array Heat Sinks by Cold Spray Additive Manufacturing: Economics of Powder Recycling. *J Therm Spray* 28:144–160
98. Stimpson CK, Snyder JC, Thole KA, Mongillo D (2018) Effectiveness Measurements of Additively Manufactured Film Cooling Holes. *J Turbomach*. doi: 10.1115/1.4038182
99. Beese AM, Carroll BE (2016) Review of Mechanical Properties of Ti-6Al-4V Made by

Laser-Based Additive Manufacturing Using Powder Feedstock. JOM 68:724–734

100. Gong H, Rafi K, Gu H, Janaki Ram GD, Starr T, Stucker B (2015) Influence of defects on mechanical properties of Ti–6Al–4 V components produced by selective laser melting and electron beam melting. *Mater Des* 86:545–554
101. Loh GH, Pei E, Harrison D, Monzón MD (2018) An overview of functionally graded additive manufacturing. *Addit Manuf* 23:34–44
102. Onuikwe B, Heer B, Bandyopadhyay A (2018) Additive manufacturing of Inconel 718—Copper alloy bimetallic structure using laser engineered net shaping (LENSTM). *Addit Manuf* 21:133–140
103. Popovich VA, Borisov E V., Heurtebise V, Riemsdijk T, Popovich AA, Sufiiarov VS (2018) Creep and thermomechanical fatigue of functionally graded inconel 718 produced by additive manufacturing. In: *Miner. Met. Mater. Ser.* Springer International Publishing, pp 85–97
104. Zocca A, Colombo P, Gomes CM, Günster J (2015) Additive Manufacturing of Ceramics: Issues, Potentialities, and Opportunities. *J Am Ceram Soc* 98:1983–2001
105. Alm B, Knitter R, Haußelt J (2005) Development of a Ceramic Micro Heat Exchanger - Design, Construction, and Testing. *Chem Eng Technol* 28:1554–1560
106. Scheithauer U, Schwarzer E, Moritz T, Michaelis A (2018) Additive Manufacturing of Ceramic Heat Exchanger: Opportunities and Limits of the Lithography-Based Ceramic Manufacturing (LCM). *J Mater Eng Perform* 27:14–20
107. Shulman H, Ross N (2015) Additive Manufacturing for Cost Efficient Production of Compact Ceramic Heat Exchangers and Recuperators. doi: 10.2172/1234436
108. Pelanconi M, Barbato M, Zavattoni S, Vignoles GL, Ortona A (2019) Thermal design, optimization and additive manufacturing of ceramic regular structures to maximize the radiative heat transfer. *Mater Des* 163:107539
109. Arie MA, Hymas DM, Singer F, Shooshtari AH, Ohadi M (2020) An additively manufactured novel polymer composite heat exchanger for dry cooling applications. *Int J Heat Mass Transf* 147:118889
110. Hymas DM, Arle MA, Singer F, Shooshtari AH, Ohadi MM (2017) Enhanced air-side heat transfer in an additively manufactured polymer composite heat exchanger. In: 2017 16th IEEE Intersoc. Conf. Therm. Thermomechanical Phenom. Electron. Syst. IEEE, pp 634–638
111. Cevallos JG, Bergles AE, Bar-Cohen A, Rodgers P, Gupta SK (2012) Polymer Heat Exchangers—History, Opportunities, and Challenges. *Heat Transf Eng* 33:1075–1093
112. Deisenroth DC, Moradi R, Shooshtari AH, Singer F, Bar-Cohen A, Ohadi M (2018) Review of Heat Exchangers Enabled by Polymer and Polymer Composite Additive Manufacturing. *Heat Transf Eng* 39:1652–1668
113. Liao H, Aldwell B, Yin S, Li W, Lupoi R, Cavaliere P, Jenkins R (2018) Cold spray

- additive manufacturing and repair: Fundamentals and applications. *Addit Manuf* 21:628–650
114. Yang K, Li W, Guo X, Yang X, Xu Y (2018) Characterizations and anisotropy of cold-spraying additive-manufactured copper bulk. *J Mater Sci Technol* 34:1570–1579
 115. Kranz J, Herzog D, Emmelmann C (2015) Design guidelines for laser additive manufacturing of lightweight structures in TiAl6V4. *J Laser Appl* 27:S14001
 116. Wiberg A, Persson J, Ölvander J Design for additive manufacturing-a review of available design methods and software. doi: 10.1108/RPJ-10-2018-0262
 117. Strano G, Hao L, Everson RM, Evans KE (2013) Surface roughness analysis, modelling and prediction in selective laser melting. *J Mater Process Technol* 213:589–597
 118. Delgado J, Ciurana J, Rodríguez CA (2012) Influence of process parameters on part quality and mechanical properties for DMLS and SLM with iron-based materials. *Int J Adv Manuf Technol* 60:601–610
 119. Yadroitsev I, Smurov I (2011) Surface Morphology in Selective Laser Melting of Metal Powders. *Phys Procedia* 12:264–270
 120. Wegner A, Witt G (2012) Correlation of Process Parameters and Part Properties in Laser Sintering using Response Surface Modeling. *Phys Procedia* 39:480–490
 121. Snyder JC, Stimpson CK, Thole KA, Mongillo D (2016) Build direction effects on additively manufactured channels. *J Turbomach* 138:051006
 122. Pakkanen J, Calignano F, Trevisan F, Lorusso M, Ambrosio EP, Manfredi D, Fino P (2016) Study of Internal Channel Surface Roughnesses Manufactured by Selective Laser Melting in Aluminum and Titanium Alloys. *Metall Mater Trans A* 47:3837–3844
 123. Bacchewar PB, Singhal SK, Pandey PM (2007) Statistical modelling and optimization of surface roughness in the selective laser sintering process. *Proc Inst Mech Eng Part B J Eng Manuf* 221:35–52
 124. Mumtaz K, Hopkinson N (2009) Top surface and side roughness of Inconel 625 parts processed using selective laser melting. *Rapid Prototyp J* 15:96–103
 125. Ramirez DA, Murr LE, Li SJ, Tian YX, Martinez E, Martinez JL, Machado BI, Gaytan SM, Medina F, Wicker RB (2011) Open-cellular copper structures fabricated by additive manufacturing using electron beam melting. *Mater Sci Eng A* 528:5379–5386
 126. Chang SW, Lees AW (2010) Endwall heat transfer and pressure drop in scale-roughened pin-fin channels. *Int J Therm Sci* 49:702–713
 127. Heo KY, Kihm KD, Lee JS (2014) Fabrication and experiment of micro-pin-finned microchannels to study surface roughness effects on convective heat transfer. *J Micromechanics Microengineering* 24:125025
 128. Guo X, Zhou J, Zhang W, Du Z, Liu C, Liu Y (2017) Self-supporting structure design in additive manufacturing through explicit topology optimization. *Comput Methods Appl Mech Eng* 323:27–63

129. Snyder JC, Stimpson CK, Thole KA, Mongillo DJ (2015) Build Direction Effects on Microchannel Tolerance and Surface Roughness. *J Mech Des.* doi: 10.1115/1.4031071
130. Collins IL, Weibel JA, Pan L, Garimella S V. (2018) Experimental Characterization of a Microchannel Heat Sink Made by Additive Manufacturing. In: 2018 17th IEEE Intersoc. Conf. Therm. Thermomechanical Phenom. Electron. Syst. IEEE, pp 171–177
131. Stimpson CK, Snyder JC, Thole KA, Mongillo D (2017) Scaling Roughness Effects on Pressure Loss and Heat Transfer of Additively Manufactured Channels. *J Turbomach.* doi: 10.1115/1.4034555
132. Saltzman D, Bichnevicius M, Lynch S, Simpson TW, Reutzel EW, Dickman C, Martukanitz R (2018) Design and evaluation of an additively manufactured aircraft heat exchanger. *Appl Therm Eng* 138:254–263
133. Collins IL, Weibel JA, Pan L, Garimella S V. (2019) A permeable-membrane microchannel heat sink made by additive manufacturing. *Int J Heat Mass Transf* 131:1174–1183
134. Stimpson CK, Snyder JC, Thole KA, Mongillo D (2018) Effects of Coolant Feed Direction on Additively Manufactured Film Cooling Holes. *J Turbomach.* doi: 10.1115/1.4041374
135. Arie MA, Shooshtari AH, Ohadi MM (2018) Experimental characterization of an additively manufactured heat exchanger for dry cooling of power plants. *Appl Therm Eng* 129:187–198
136. Arie MA, Shooshtari A, Ohadi M (2017) Air side enhancement of heat transfer in an additively manufactured 1 kW heat exchanger for dry cooling applications. In: 2017 16th IEEE Intersoc. Conf. Therm. Thermomechanical Phenom. Electron. Syst. IEEE, pp 1–8